

SaaS Product Analytics Case Study

SaaS Behavior • Revenue • Retention Analysis using SQL



Built using PostgreSQL • SQL • Excel

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ABSTRACT

This project analyses a simulated SaaS subscription platform dataset to study customer lifecycle behaviour across revenue, feature engagement, support interactions, retention, and churn. Using PostgreSQL and relational data modelling, the analysis explores how user activity, subscription movement, product adoption, and support experience influence business growth and long-term customer retention across the platform.

The goal of this analysis is to identify behavioural patterns such as engagement intensity, beta feature adoption, and product friction, and evaluate how these factors influence upgrades, downgrades, churn, and overall platform growth.

Dataset Overview

Contents of Tables

Syntax: **Select * FROM tablename;**

1. Accounts :

	account_id text 🔒	account_name text 🔒	industry text 🔒	country text 🔒	signup_date date 🔒	referral_source text 🔒	plan_tier text 🔒	seats integer 🔒	is_trial text 🔒	churn_flag text 🔒
1	A-2e4581	Company_0	EdTech	US	2024-10-16	partner	Basic	9	False	False
2	A-43a9e3	Company_1	FinTech	IN	2023-08-17	other	Basic	18	False	True
3	A-0a282f	Company_2	DevTools	US	2024-08-27	organic	Basic	1	False	False
4	A-1f0ac7	Company_3	HealthTe...	UK	2023-08-27	other	Basic	24	True	False
5	A-ce550d	Company_4	HealthTe...	US	2024-10-27	event	Enterprise	35	False	True

500 rows

2. Subscriptions :

	subscription_id text	account_id text	start_date date	end_date date	plan_tier text	seats integer	mrr_amount numeric	arr_amount numeric	is_trial text	upgrade_flag text	downgrade_flag text	churn_flag text	billing_frequency text
1	S-8cec59	A-3c1a3f	2023-12-23	2024-04-12	Enterprise	14	2786	33432	False	False	False	True	monthly
2	S-0f6f44	A-9b9fe9	2024-06-11	[null]	Pro	17	833	9996	False	False	False	False	monthly
3	S-51c0d1	A-659280	2024-11-25	[null]	Enterprise	62	0	0	True	True	False	False	annual
4	S-f81687	A-e7a1e2	2024-11-23	2024-12-13	Enterprise	5	995	11940	False	False	False	True	monthly
5	S-cff5a2	A-ba6516	2024-01-10	[null]	Enterprise	27	5373	64476	False	False	False	False	monthly

5000 rows

3. Feature_usage

	usage_id text	subscription_id text	usage_date text	feature_name text	usage_count text	usage_duration_secs text	error_count text	is_beta_feature text
1	U-1c6c24	S-0fcf7d	2023-07-27	feature_20	9	5004	0	False
2	U-f07cb8	S-c25263	2023-08-07	feature_5	9	369	0	False
3	U-096807	S-f29e7f	2023-12-07	feature_3	9	1458	0	False
4	U-6b1580	S-be655e	2024-07-28	feature_40	5	2085	0	False
5	U-720a29	S-f9b1d0	2024-12-02	feature_12	12	900	0	False

25000 rows

4. Support_tickets:

key		ticket_id text	account_id text	submitted_at text	closed_at text	resolution_time_hours text	priority text	first_response_time_minutes text	satisfaction text
	1	T-0024de	A-712f1c	2023-07-27	2023-07-28 03:00:...	27.0	high	74	[null]
	2	T-4d04b9	A-e43bf7	2024-07-08	2024-07-09 03:00:...	27.0	urgent	144	[null]
	3	T-d5e12f	A-0f3e88	2024-10-17	2024-10-17 19:00:...	19.0	urgent	93	4.0
	4	T-dfce9a	A-4c56c9	2024-09-08	2024-09-09 23:00:...	47.0	medium	126	5.0
	5	T-c59f77	A-6f8ad2	2024-11-30	2024-12-01 02:00:...	26.0	medium	8	[null]

2000 rows

5. Churn_events

	churn_event_id text	account_id text	churn_date text	reason_code text	refund_amount_usd text	preceding_upgrade_flag text	preceding_downgrade_flag text	is_reactivation text	feedback_text text
1	C-816288	A-c37cab	2024-10-27	pricing	4.03	False	False	False	switched to competi...
2	C-5a81e7	A-37f969	2024-06-25	support	96.45	True	False	False	[null]
3	C-a174be	A-b07346	2024-11-12	budget	0.0	False	False	False	missing features
4	C-accb39	A-1e50e0	2023-11-01	budget	54.94	False	False	False	switched to competi...
5	C-92f889	A-956988	2024-12-30	unknown	0.0	False	True	True	too expensive

600 rows

SECTION 1: REVENUE & MONETIZATION

1. Revenue Contribution by Plan Tier

Which subscription plan tiers generate the highest recurring revenue, and how does revenue distribution vary across tiers?

```
SELECT
  plan_tier,
  COUNT(DISTINCT subscription_id) AS total_subscriptions,
  SUM(mrr_amount) AS total_mrr,
  SUM(arr_amount) AS total_arr,
  ROUND(AVG(mrr_amount), 2) AS avg_mrr_per_subscription
FROM subscriptions
WHERE is_trial = 'False'
GROUP BY plan_tier
ORDER BY total_mrr DESC;
```

	plan_tier 	total_subscriptions 	total_mrr 	total_arr 	avg_mrr_per_subscription 
1	Enterprise	1450	8473221	101678652	5843.60
2	Pro	1416	2105089	25261068	1486.64
3	Basic	1356	760437	9125244	560.79

2. Subscription Tier Movement (Upgrades vs Downgrades)

How do users transition between subscription tiers through upgrades and downgrades, and what does this indicate about product growth?

```
SELECT
  plan_tier,
  SUM(CASE WHEN upgrade_flag = 'True' THEN 1 ELSE 0 END) AS upgrades,
  SUM(CASE WHEN downgrade_flag = 'True' THEN 1 ELSE 0 END) AS downgrades,
  COUNT(subscription_id) AS total_subscriptions
FROM subscriptions
GROUP BY plan_tier
ORDER BY upgrades DESC;
```

	plan_tier 	upgrades 	downgrades 	total_subscriptions 
1	Enterprise	194	68	1723
2	Pro	194	90	1675
3	Basic	141	60	1602

Users show a higher tendency to upgrade than downgrade, suggesting healthy upward movement across subscription tiers and overall product adoption growth.

3. Revenue Impact of Upgrades and Downgrades

How do upgrades and downgrades impact net revenue across subscription tiers?

```
SELECT
  plan_tier,
  SUM(CASE WHEN upgrade_flag = 'True' THEN mrr_amount ELSE 0 END) AS upgrade_revenue,
  SUM(CASE WHEN downgrade_flag = 'True' THEN mrr_amount ELSE 0 END) AS downgrade_revenue,
  SUM(CASE
    WHEN upgrade_flag = 'True' THEN mrr_amount
    WHEN downgrade_flag = 'True' THEN -mrr_amount
    ELSE 0
  END) AS net_revenue_impact
FROM subscriptions
GROUP BY plan_tier
ORDER BY net_revenue_impact DESC;
```

	plan_tier 	upgrade_revenue 	downgrade_revenue 	net_revenue_impact 
	text	numeric	numeric	numeric
1	Enterprise	921768	318002	630233
2	Pro	281113	114023	178017
3	Basic	60116	27341	35815

Revenue impact is mainly driven by upgrades, while downgrades act as controlled leakage, showing that expansion revenue plays a stronger role than contraction across subscription tiers.

SECTION 2: CUSTOMER ACQUISITION & CONVERSION

4. Trial vs Paid Subscription Distribution

How effectively do trial users convert into paid subscriptions across different plan tiers?

```
SELECT
  plan_tier,
  COUNT(subscription_id) AS total_subscriptions,
  SUM(CASE WHEN is_trial = 'True' THEN 1 ELSE 0 END) AS trial_users,
  SUM(CASE WHEN is_trial = 'False' THEN 1 ELSE 0 END) AS paid_users,
  ROUND(
    SUM(CASE WHEN is_trial = 'False' THEN 1 ELSE 0 END) * 100.0
    / COUNT(subscription_id), 2
  ) AS paid_conversion_rate
FROM subscriptions
GROUP BY plan_tier
ORDER BY paid_conversion_rate DESC;
```

	plan_tier 	total_subscriptions 	trial_users 	paid_users 	paid_conversion_rate 
	text	bigint	bigint	bigint	numeric
1	Basic	1602	246	1356	84.64
2	Pro	1675	259	1416	84.54
3	Enterprise	1723	273	1450	84.16

Paid conversion rates remain relatively consistent across plan tiers, suggesting that trial-to-paid behaviour is fairly stable throughout the product ecosystem.

SECTION 3: PRODUCT ENGAGEMENT

5. Feature Usage Distribution

Which product features drive the highest user engagement?

```
SELECT
  f.feature_name,
  COUNT(DISTINCT f.subscription_id) AS total_users,
  SUM(f.usage_count) AS total_feature_usage,
  ROUND(AVG(f.usage_duration_secs) / 60.0, 2) AS avg_usage_minutes,
  SUM(CASE
    WHEN f.is_beta_feature = 'True' THEN 1
    ELSE 0
  END) AS beta_feature_usage,
  SUM(CASE
    WHEN s.upgrade_flag = 'True' THEN 1
    ELSE 0
  END) AS upgraded_users,
  SUM(CASE
    WHEN s.churn_flag = 'True' THEN 1
    ELSE 0
  END) AS churned_users
FROM feature_usage f
JOIN subscriptions s
  ON f.subscription_id = s.subscription_id
GROUP BY f.feature_name
ORDER BY total_feature_usage DESC;
```

	feature_name 	total_users 	total_feature_usage 	avg_usage_minutes 	beta_feature_usage 	upgraded_users 	churned_users 
1	feature_32	614	34147	51.90	67	60	65
2	feature_12	624	34036	52.14	65	84	69
3	feature_6	604	33938	51.25	71	68	69
4	feature_36	604	33791	48.22	63	67	70
5	feature_17	613	33774	49.94	60	74	46
6	feature_34	613	33749	51.45	67	72	52
7	feature_26	616	33614	50.75	63	74	51
8	feature_24	601	33416	53.66	69	59	50
9	feature_31	596	33371	47.48	62	52	59
10	feature_2	607	33286	50.25	69	80	73

40 rows

Some high-usage features also show churn signals, suggesting possible gaps in usability or perceived value.

6. Beta Feature Engagement vs Non-Beta Users

“Do users who engage with beta features demonstrate stronger engagement and retention behaviour?”

```
SELECT
  CASE
    WHEN f.is_beta_feature = 'True' THEN 'Beta Users'
    ELSE 'Non-Beta Users'
  END AS user_segment,
  COUNT(DISTINCT f.subscription_id) AS total_users,
  ROUND(AVG(f.usage_count), 2) AS avg_feature_usage,
  ROUND(AVG(f.usage_duration_secs) / 60, 2) AS avg_usage_minutes,
  SUM(CASE
    WHEN s.upgrade_flag = 'True' THEN 1
    ELSE 0
  END) AS upgraded_users,
  SUM(CASE
    WHEN s.churn_flag = 'True' THEN 1
    ELSE 0
  END) AS churned_users,
  ROUND(
    (
      SUM(CASE
        WHEN s.churn_flag = 'True' THEN 1
        ELSE 0
      END) * 100.0
    ) / COUNT(DISTINCT f.subscription_id),
    2
  ) AS churn_rate
FROM feature_usage f
JOIN subscriptions s
  ON f.subscription_id = s.subscription_id
GROUP BY user_segment;
```

	user_segment 	total_users 	avg_feature_usage 	avg_usage_minutes 	upgraded_users 	churned_users 	churn_rate 
1	Beta Users	2023	51.78	50.82	270	240	11.86
2	Non-Beta Users	4946	51.86	50.69	2403	2133	43.13

Users engaging with beta features demonstrate significantly lower churn rates and stronger upgrade behaviour compared to non-beta users, suggesting that advanced feature adoption is strongly associated with long-term retention and monetization potential.

7(a). User Engagement vs Upgrade / Downgrade / Churn Behavior

“Do users who engage with beta features demonstrate stronger engagement and retention behaviour?”

```
SELECT
  s.subscription_id,
  s.plan_tier,
  SUM(f.usage_count) AS total_usage,
  ROUND(SUM(f.usage_duration_secs) / 60.0, 2) AS total_time_minutes,
  SUM(f.error_count) AS total_errors,
  SUM(CASE WHEN s.upgrade_flag = 'True' THEN 1 ELSE 0 END) AS upgrades,
  SUM(CASE WHEN s.downgrade_flag = 'True' THEN 1 ELSE 0 END) AS downgrades,
  SUM(CASE WHEN s.churn_flag = 'True' THEN 1 ELSE 0 END) AS churns
FROM subscriptions s
LEFT JOIN feature_usage f
  ON s.subscription_id = f.subscription_id
GROUP BY s.subscription_id, s.plan_tier;
```

This query aggregates subscription-level behavioural signals by combining usage intensity, error patterns, and lifecycle events like upgrade and churn to build a unified customer health profile.

	subscription_id	plan_tier	total_usage	total_time_minutes	total_errors	upgrades	downgrades	churns
	text	text	bigint	numeric	bigint	bigint	bigint	bigint
1	S-17cfc4	Pro	470	447.22	2	0	9	0
2	S-fe753c	Basic	218	175.28	2	0	0	0
3	S-f52fd6	Enterprise	257	271.07	8	0	0	0
4	S-43af5a	Pro	221	161.45	1	0	0	0
5	S-a0a5e3	Enterprise	366	450.15	1	0	0	0
6	S-c284ce	Enterprise	270	150.12	0	0	0	0
7	S-f4097a	Enterprise	330	141.83	7	0	0	0
8	S-765a5f	Pro	260	490.82	3	0	0	5
9	S-22083e	Pro	420	437.50	3	0	0	0
10	S-615a8a	Enterprise	310	163.33	3	0	0	0

5000 rows

7(b). User Engagement vs Upgrade / Downgrade / Churn Behaviour

How do engagement intensity and product friction together influence customer lifecycle outcomes such as upgrades, downgrades, and churn?

```
SELECT
CASE
  WHEN f.usage_count <= 20 THEN 'Low Usage'
  WHEN f.usage_count BETWEEN 21 AND 60 THEN 'Medium Usage'
  ELSE 'High Usage'
END AS usage_segment,
CASE
  WHEN f.error_count = 0 THEN 'No Errors'
  WHEN f.error_count BETWEEN 1 AND 3 THEN 'Low Errors'
  ELSE 'High Errors'
END AS error_segment,
COUNT(DISTINCT s.subscription_id) AS total_users,
SUM(CASE WHEN s.upgrade_flag = 'True' THEN 1 ELSE 0 END) AS upgrades,
SUM(CASE WHEN s.downgrade_flag = 'True' THEN 1 ELSE 0 END) AS downgrades,
SUM(CASE WHEN s.churn_flag = 'True' THEN 1 ELSE 0 END) AS churns
FROM subscriptions s
LEFT JOIN feature_usage f
  ON s.subscription_id = f.subscription_id
GROUP BY usage_segment, error_segment
ORDER BY upgrades DESC;
```

	usage_segment text	error_segment text	total_users bigint	upgrades bigint	downgrades bigint	churns bigint
1	Medium Usage	No Errors	4848	1876	741	1627
2	Medium Usage	Low Errors	3790	742	321	688
3	Medium Usage	High Errors	536	55	36	58
4	High Usage	High Errors	33	4	2	3

Higher usage intensity does not always guarantee retention; instead, the combination of usage and error patterns better explains upgrades, downgrades, and churn behavior.

SECTION 4: CUSTOMER SUPPORT & EXPERIENCE ANALYSIS

8. Support Load & Satisfaction Across Plan Tiers and Churn Status

How does customer support usage and satisfaction vary across plan tiers and churn behavior?

```
SELECT
  s.plan_tier,
  s.churn_flag,
  COUNT(st.ticket_id) AS total_tickets,
  ROUND(AVG(st.resolution_time_hours::numeric), 2) AS avg_resolution_time_hours,
  ROUND(AVG(st.satisfaction_score::numeric), 2) AS avg_satisfaction_score
FROM subscriptions s
LEFT JOIN support_tickets st
  ON s.account_id = st.account_id
GROUP BY s.plan_tier, s.churn_flag
ORDER BY s.plan_tier;
```

	plan_tier 	churn_flag 	total_tickets 	avg_resolution_time_hours 	avg_satisfaction_score 
1	Basic	True	590	35.52	3.98
2	Basic	False	5810	35.89	4.00
3	Enterprise	True	698	37.46	3.97
4	Enterprise	False	6116	36.18	3.99
5	Pro	False	6002	35.88	3.99
6	Pro	True	682	35.15	3.96

Support experience alone may not be the primary driver of churn in this dataset. Churned and active users show nearly identical usage and error patterns. This indicates that churn is not driven by engagement volume or system errors, but likely by deeper factors such as perceived value, pricing sensitivity, or unmet product expectations.

SECTION 5: RETENTION & GROWTH BEHAVIOR

9. Cohort-Based User Activity Over Time

How do user cohorts behave over time in terms of product activity and retention patterns?

```
WITH active_months AS (  
  SELECT DISTINCT  
    a.account_id,  
    TO_CHAR(DATE_TRUNC('month', a.signup_date::date), 'YYYY-MM') AS cohort_month,  
    TO_CHAR(DATE_TRUNC('month', f.usage_date::date), 'YYYY-MM') AS activity_month  
  FROM accounts a  
  JOIN subscriptions s  
    ON a.account_id = s.account_id  
  JOIN feature_usage f  
    ON s.subscription_id = f.subscription_id  
  WHERE f.usage_date::date >= a.signup_date::date  
)  
SELECT  
  cohort_month,  
  activity_month,  
  COUNT(DISTINCT account_id) AS active_users  
FROM active_months  
GROUP BY cohort_month, activity_month  
ORDER BY cohort_month, activity_month;
```

	cohort_month 	activity_month 	active_users 
	text	text	bigint
1	2023-01	2023-01	8
2	2023-01	2023-02	12
3	2023-01	2023-03	15
4	2023-01	2023-04	16
5	2023-01	2023-05	16
6	2023-01	2023-06	13
7	2023-01	2023-07	16
8	2023-01	2023-08	17
9	2023-01	2023-09	14
10	2023-01	2023-10	15

300 rows

User activity generally increases across later activity periods, suggesting steady platform adoption and continued engagement growth over time.

10. Monthly Active Users & Engagement Trends

How is overall platform engagement trending over time in terms of active users and usage intensity?

```
SELECT
  TO_CHAR(Date_TRUNC('month', f.usage_date::date), 'YYYY-MM') AS activity_month,
  COUNT(DISTINCT s.account_id) AS monthly_active_users,
  COUNT(DISTINCT f.subscription_id) AS active_subscriptions,
  SUM(f.usage_count) AS total_platform_usage,
  ROUND(AVG(f.usage_duration_secs) / 60, 2) AS avg_usage_minutes
FROM feature_usage f
JOIN subscriptions s
  ON f.subscription_id = s.subscription_id
GROUP BY activity_month
```

	activity_month text	monthly_active_users bigint	active_subscriptions bigint	total_platform_usage bigint	avg_usage_minutes numeric
1	2023-01	440	969	55344	52.47
2	2023-02	412	875	50363	49.25
3	2023-03	431	989	57228	50.70
4	2023-04	432	932	53416	52.88
5	2023-05	438	978	55852	50.04
6	2023-06	420	907	52758	50.18
7	2023-07	415	960	55628	51.10
8	2023-08	426	953	54026	50.20
9	2023-09	409	878	50212	50.90
10	2023-10	426	971	56785	51.10

24 rows

Overall engagement trends show stable active usage with fluctuations in intensity, suggesting consistent but non-linear platform growth.

11. DAU/MAU Stickiness Ratio

How frequently do users return to the platform on a daily basis (stickiness)?

```
WITH daily_activity AS (  
    SELECT  
        usage_date,  
        COUNT(DISTINCT subscription_id) AS dau  
    FROM feature_usage  
    GROUP BY usage_date  
),  
monthly_activity AS (  
    SELECT  
        DATE_TRUNC('month', usage_date) AS activity_month,  
        COUNT(DISTINCT subscription_id) AS mau  
    FROM feature_usage  
    GROUP BY activity_month  
),  
monthly_dau AS (  
    SELECT  
        DATE_TRUNC('month', usage_date) AS activity_month,  
        AVG(dau) AS avg_daily_active_users  
    FROM daily_activity  
    GROUP BY activity_month  
)  
SELECT  
    m.activity_month,  
    d.avg_daily_active_users,  
    m.mau,  
    ROUND((d.avg_daily_active_users * 100.0) / m.mau, 2) AS stickiness_ratio  
FROM monthly_activity m  
JOIN monthly_dau d  
    ON m.activity_month = d.activity_month  
ORDER BY m.activity_month;
```

	activity_month text	avg_daily_active_users numeric	mau bigint	stickiness_ratio numeric
1	2023-01	34	969	3.51
2	2023-02	35	875	4.00
3	2023-03	36	989	3.64
4	2023-04	34	932	3.65
5	2023-05	35	978	3.58
6	2023-06	34	907	3.75
7	2023-07	35	960	3.65
8	2023-08	33	953	3.46
9	2023-09	32	878	3.64
10	2023-10	35	971	3.60
11	2023-11	31	868	3.57
12	2023-12	34	945	3.60
13	2024-01	34	949	3.58
14	2024-02	35	931	3.76

Stickiness ratios indicate moderate-to-strong engagement behaviour, showing that a meaningful portion of users return frequently within each month.

SECTION 6: CHURN ANALYSIS

12. Churn Reason Breakdown

What are the primary reasons customers churn, and how does revenue loss vary by reason?

```
SELECT
  reason_code,
  COUNT(churn_event_id) AS total_churn_events,
  AVG(refund_amount_usd::numeric) AS avg_refund,
  SUM(CASE WHEN preceding_upgrade_flag = 'True' THEN 1 ELSE 0 END) AS churn_after_upgrade,
  SUM(CASE WHEN preceding_downgrade_flag = 'True' THEN 1 ELSE 0 END) AS churn_after_downgrade,
  SUM(CASE WHEN is_reactivation = 'True' THEN 1 ELSE 0 END) AS reactivated_accounts
FROM churn_events
GROUP BY reason_code
ORDER BY total_churn_events DESC;
```

	reason_code text	total_churn_events bigint	avg_refund numeric	churn_after_upgrade bigint	churn_after_downgrade bigint	reactivated_accounts bigint
1	features	114	16.7184210526315789	17	9	13
2	support	104	11.7273076923076923	28	13	6
3	budget	104	11.9955769230769231	24	6	12
4	unknown	95	18.3409473684210526	19	9	9
5	competitor	92	13.0818478260869565	17	12	12
6	pricing	91	14.6510989010989011	18	4	9

Churn is driven by a mix of pricing, usability, and unmet expectations, with some users even churning after upgrades, indicating possible expectation gaps. While several churn reasons are identifiable, a significant portion falls under “Unknown,” highlighting the limitations of churn attribution and the need for better customer feedback capture.

13. Churned vs Active User Behaviour Comparison

How does product engagement differ between churned and active users?

```
SELECT
  CASE
    WHEN s.churn_flag = TRUE THEN 'Churned'
    ELSE 'Active'
  END AS account_status,
  COUNT(DISTINCT s.subscription_id) AS total_subscriptions,
  AVG(f.usage_count) AS avg_feature_usage,
  AVG(f.error_count) AS avg_error_count
FROM subscriptions s
LEFT JOIN feature_usage f
  ON s.subscription_id = f.subscription_id
GROUP BY account_status;
```

	account_status 	total_subscriptions 	avg_feature_usage 	avg_error_count 
1	Active	4514	51.85	0.56
2	Churned	486	51.87	0.57

Since active subscriptions make up the majority of the dataset, they naturally account for a larger share of support interactions and product activity. Churned and active users show very similar engagement patterns, suggesting that churn may be influenced more by perceived value, pricing, or external factors than by raw product usage alone.

Key Business Insights

- Enterprise plans generated the highest revenue and subscription share.
- Beta feature users showed stronger retention behaviour.
- Trial-to-paid conversion remained stable across plan tiers.
- Support metrics alone did not strongly explain churn.
- Churned and active users displayed similar engagement patterns.
- Retention appears influenced by perceived value beyond raw activity levels

Tools & Skills Used

Tools	Skills	
• <u>PostgreSQL</u> • <u>Microsoft PowerPoint</u> • <u>Microsoft Excel</u> • <u>pgAdmin</u>	• SQL Querying • Joins & Relational Modeling • Aggregations & Grouping • CASE Statements • Common Table Expressions (CTEs) • Data Cleaning & Type Casting • Cohort Analysis • DAU / MAU Stickiness Analysis	• Churn Analysis • Subscription Revenue Analysis • Customer Segmentation • Behavioral Analytics • Product Usage Analysis • Retention Analysis • KPI Interpretation • Saas Metrics (ARR, MRR)